

Delhi Metro Operations Optimization Final Report

Introduction

This report analyzes the operational data of the Delhi Metro to identify key areas for optimizing train frequencies, reducing overcrowding, and improving overall service levels. By leveraging available datasets, including stop times, stop locations, trips, and routes, we have generated visualizations to highlight patterns in metro usage throughout the day and across different routes. The findings indicate distinct peak and off-peak hours, as well as variations in ridership across different metro lines. Based on these insights, we propose targeted adjustments to train frequencies to better match supply with demand.

Data Sources and Methodology

This analysis utilizes publicly available datasets from the **Delhi Metro Rail Corporation**.

stop_times.csv: Contains timetables for each trip, detailing arrival and departure times at specific stops.

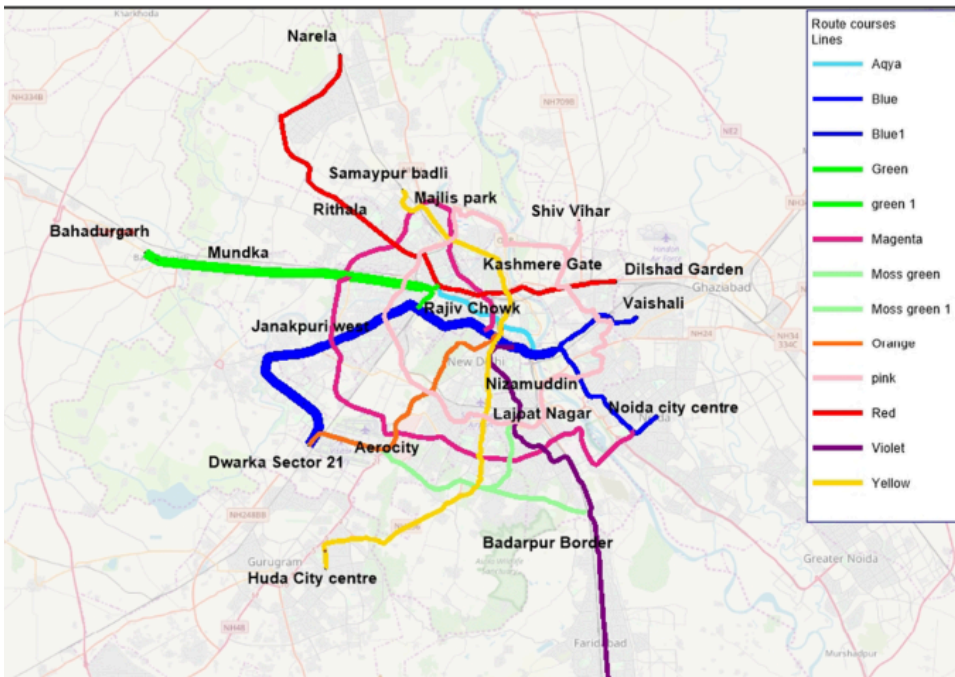
stops.csv: Provides geographical locations of metro stops, including latitude and longitude coordinates.

Trips.csv: Links individual trips to their respective routes, providing trip identifiers and associated route IDs.

Routes.csv: Contains details of metro routes, including short and long names, and route type.

Key Findings and Analysis

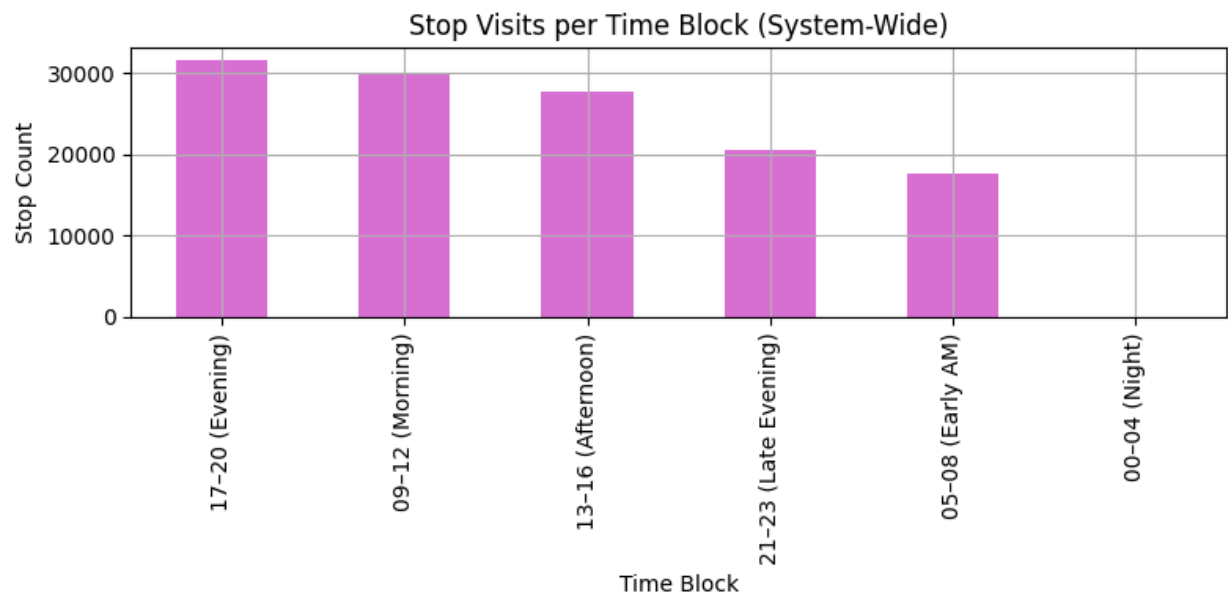
Route Map Analysis:



Geospatial plots show long arterial lines (**Blue, Yellow**) span wide distances and connect transfer stations

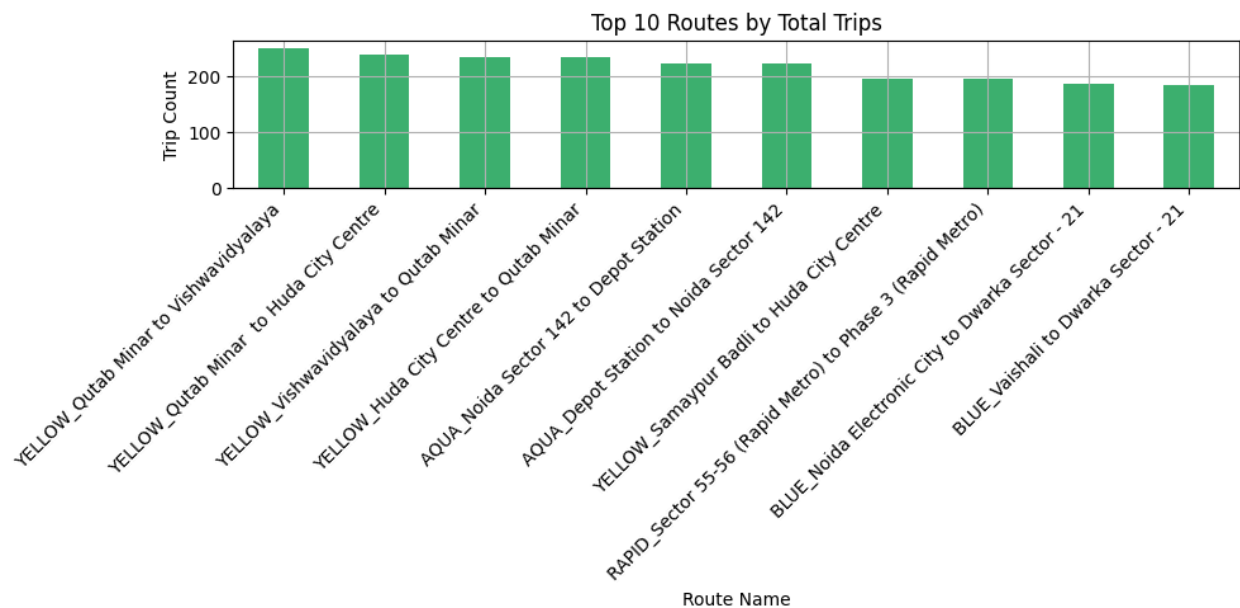
Hourly Demand Distribution:

To understand the temporal demand patterns, we analyzed the distribution of metro arrivals across different hours of the day.



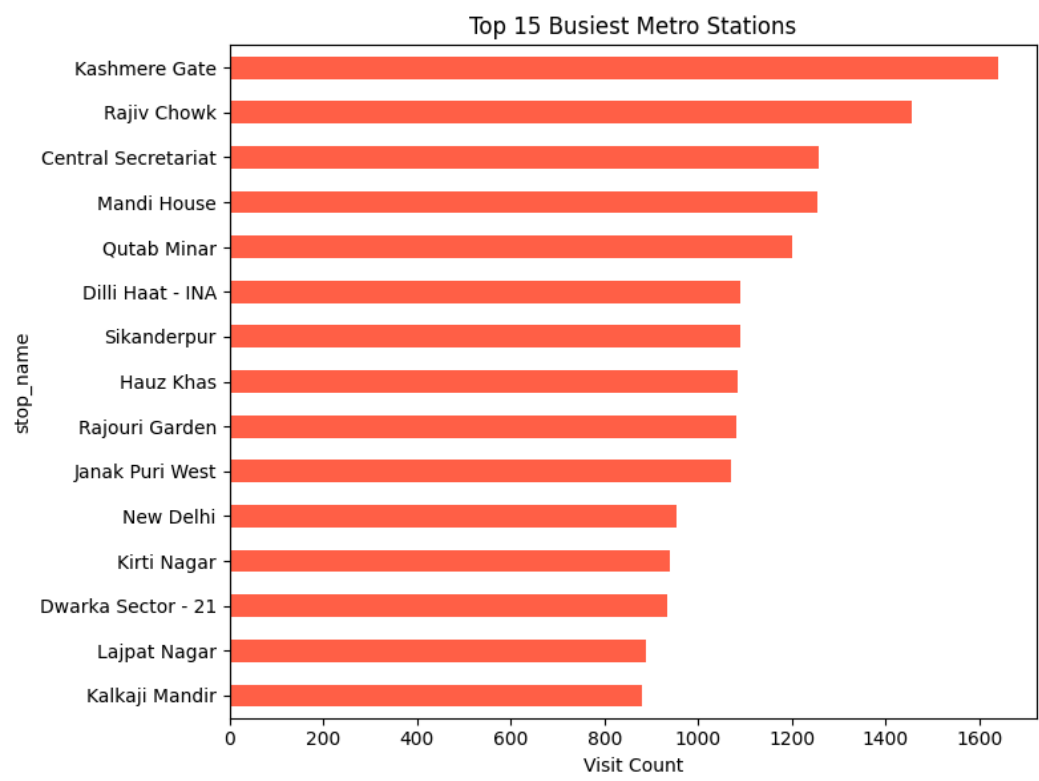
- Clear **morning peak (9–12 AM)** and **evening peak (5–8 PM)**
- Demand drops significantly late night and early morning (after 10 PM, before 6 AM)

Top 10 routes having highest trip round



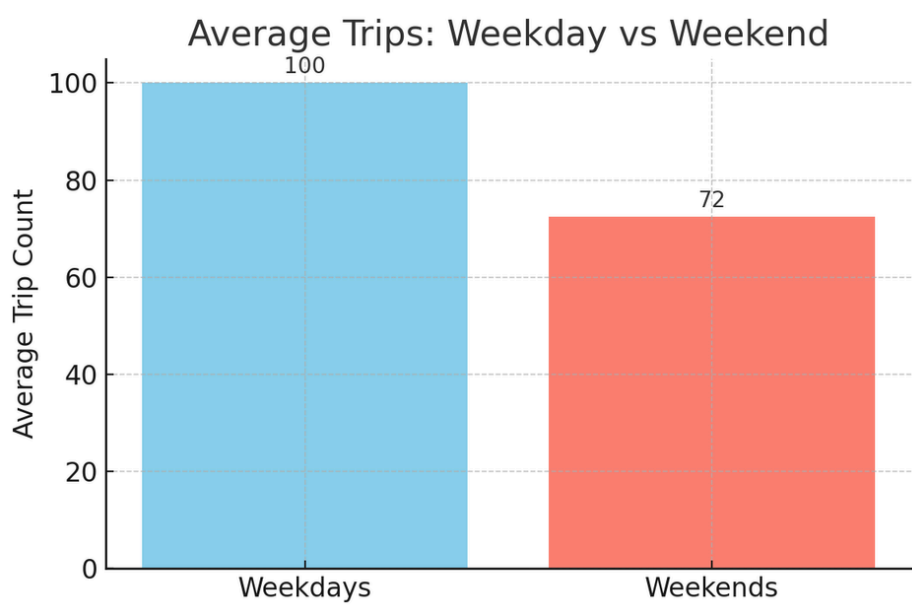
- The majority of the top 10 busiest routes belong to the **Yellow** and **Blue** metro lines.
- **"YELLOW Qutab Minar to Vishwavidyalaya"** is the most traveled route, indicating a high volume of daily commuters on this corridor.

Busiest metro station



Kashmere Gate and **Rajiv Chowk** are the busiest metro stations, indicating they serve as major interchange or high-traffic hubs.

Weekday vs Weekend Trends:



The bar graph showing that the number of metro trips is approximately **27.5% lower on weekends** compared to weekdays, highlighting reduced commuter traffic on weekends.

Recommendations for Optimization

The following recommendations are proposed to optimize train frequencies, alleviate overcrowding, and enhance overall service levels:

1. Dynamic Frequency Adjustments Based on Hourly Demand

The analysis of metro arrivals by hour of day clearly indicates **significant fluctuations** in demand throughout the day. To address this, a dynamic frequency adjustment strategy is recommended:

- **Increase Train Frequencies During Peak Hours:**

During the identified **morning peak** (9AM - 12AM) and **evening peak** (5PM -8 PM), train frequencies should be **significantly increased**. This will help distribute passenger load **more evenly across** trains, **reduce platform crowding**, and **decrease wait times** for commuters. Implementing **shorter time between trains** during these periods will be crucial.

- **Optimize Frequencies During Off-Peak Hours:**

In contrast, during **off-peak hours** (early morning before 7 AM and late evening after 9 PM), frequencies can be **moderately reduced**. This optimization will prevent the running of **underutilized trains**, leading to **energy savings** and reduced operational costs without significantly impacting the commuter experience during periods of lower demand. The saved resources can then be reallocated to support increased frequencies during peak times.

2.Route-Specific Service Enhancements

The varying number of trips across different metro routes highlights the need for route specific optimization strategies:

- **Prioritize High-Demand Routes:**

The '**Red**', '**Blue**', and '**Green**' routes, identified as the **busiest**, should be prioritized for service enhancements. This includes not only increasing frequencies during peak hours but also potentially **deploying longer trains** or trains with higher carrying capacity on these lines.

- **Strategic Review of Lower-Demand Routes:**

For routes with **fewer trips**, such as '**Light Blue**' and '**Maroon**', a detailed review of their **operational schedules** and passenger demand patterns is recommended. It is important to assess if these routes serve specific populations or **connect to critical hubs** that might experience localized peak demands at different times. Adjustments should aim to maintain a balance between operational efficiency and adequate service provision for the communities served by these.

3. Leveraging Technology for Real-time Optimization

To implement dynamic frequency adjustments effectively, the Delhi Metro can **leverage advanced technologies**:

- **Real-time Passenger Monitoring Systems:**

Deploying or enhancing systems that provide **real-time data on passenger** numbers at stations and inside trains can enable immediate adjustments to train dispatching, and alter frequencies in response to **real-time demand** and operational incidents and routing. This could involve using sensor data, CCTV analytics, or even anonymized mobile network data to gauge crowding levels.

- **Predictive Analytics for Demand Forecasting:**

Utilizing historical data combined with **machine learning models to predict future demand** fluctuations more accurately. This would allow for proactive scheduling adjustments rather than reactive responses to overcrowding.

4. Dwell time and Dynamic Scheduling

- **Adjust Station Dwell Time**

Interchange stations like Rajiv Chowk, Kashmiri Gate, and Central Secretariat experience very high footfall. If the train halts for too little time:

- ☐ Not all passengers can board or exit
- ☐ Passengers may push or rush, causing delays or even safety risks
- ☐ Trains get delayed downstream due to longer boarding/alighting

Increase dwell time specifically at high-demand interchange stations (e.g., from 30 sec to 45–60 sec).

- **Dynamic Scheduling Using Calendar Data**

On **weekends** and holidays, passenger flow is **usually lower**. Running a full weekday-level service wastes resources like higher electricity, crew costs and underutilised trains.

Use calendar.txt to identify weekday and weekend service patterns. **Run fewer** trains or **reduce frequency** on weekends.

By implementing these recommendations, the Delhi Metro can significantly improve its operational efficiency, enhance commuter satisfaction, and solidify its role as the backbone of public transportation in the National Capital Region. These strategies will contribute to a more sustainable, accessible, and user-friendly urban transit system.

